

2.1 Recording Media

To understand the progression towards digital video, one needs to understand a bit about the recording quality of various formats past and present. The quality of video is typically measured in *horizontal lines of resolution*. Each picture frame is made up of a number of lines of horizontal lines that “draw” the details of the picture. The simple arithmetic is that as the number of horizontal lines of resolution increases, so does picture quality. With more number of lines comes more detail in every aspect of the picture from skin colour, strands of hair, and other fine-grained details—which actually means every aspect of the picture. TV broadcasts are typically at around 330 lines of resolution. VHS tapes play back at around 240 to 270 lines. The digital formats reviewed here all support up to 500 lines of resolution or higher, depending on the media type and the camcorder.

In the days of the VCR, there was a format war for the VCR tape: Betamax vs. VHS. As you know, VHS won that war. In the case of camcorders, though, history is not repeating itself. The formats jungle has with no clear winners yet. To ease the confusion, we’ve categorised this discussion according to media type: tape, discs, and flash memory.

2.1.1. Tape

Tape used to be associated with analogue to digital conversion and the resulting loss of quality. No longer. Digital camcorders use digital recording to store data on digital tapes—this ensures there is no loss of video information.

In analogue camcorders, the most popular recording format used to be 8mm/Hi8 tapes. The digital equivalent is the Digital8 format, which delivers high-quality 500-line resolution clarity, depending on the camcorder. Digital8 is a bit larger in size compared to MiniDV (which we’ll soon talk about), but it is also cheaper. The tape size also has an impact on the form factor, which limits how compact the camcorder can get. Digital8-based camcorders